Sequences — problem 9.42

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A sequence is defined as follows : $a_1 = 3$, and

$$a_{n+1} = \begin{cases} a_n/2 & \text{if } a_n \text{ is even,} \\ (a_n + 1983)/2 & \text{if } a_n \text{ is odd.} \end{cases}$$

Prove that it is periodic and find its minimal period.